

Gibb's reflective cycle

1. Description

What happened?

I compared three public datasets against the task sheet rubric, read the Lichess release notes and known issues documentation, and drafted a three-tier lakehouse architecture with Star-schema Gold layer. I mapped each pipeline stage to what/why/who/how framework specified in Exercise 1(Step 2) and documented the storage plan trade-offs across OBT, star, snowflake, and 3NF structures.

2. Feelings

What were you thinking and feeling?

Initially overwhelmed by the terabyte-scale nature of the Lichess dataset and the breadth of pipeline stages required. Reassured by the CC0 licensing, the monthly file structure enabling a phased approach, and the availability of well-documented tools(python-chess, Delta Lake, MLFlow) that reduce implementation risk. The mapping between pipeline stages and business consumers (analysts vs ML pipeline) clarified the architecture choices.

3. Evaluation

What was good and bad about the experience?

The star schema and Delta Lake Silver layer are well-justified for the BI and ML consumer requirements. The three-tier lakehouse cleanly separates concerns. A weakness in this version is that the player dimension's slowly changing nature (payer Elo evolves over time) is acknowledged but not yet fully designed. SCD2 logic is referenced but not detailed. The monitoring stage is planned but the specific drift thresholds and alert policies are not yet defined.

4. Analysis

What sense can you make of a situation?

The highest risks are (1) PGN parse edge cases introduced by schema changes across 159 monthly files mitigated by the bronze immutability principle and automated QA in Silver, (2) data drift between training and serving distributions as the Lichess player serving distributions as the Lichess player pool evolves mitigated by monthly retraining aligned to the release cadence. Automated QA and monitoring are not optional components

6. Action plan

If it arose again, What would you do?

- Prototype the bronze-to-silver path on a 1 million game slice to validate the parser and schema assumptions.
- Register a minimal LightGBM outcome model with MLFlow to test the registry gate logic.
- Add unit tests for the PGN parser.
- Consult with the module tutor on whether the first model iteration should remain outcome-only before adding eval-gated move features.

5. Conclusion

What else could you have done?

This v1 plan adds explicit monitoring and deployment gates that were absent in the initial component sketch (v0). The star schema choice is now better justified against alternatives, and the ELT process is more precisely specified.